Single-Tag Range–Inertial Position Acquisition with Profiled Heading and a Rotating Bias Model

Abstract— **An aircraft with a single ultra-wideband tag receives distances to different anchors at different times. Given gravity alignment, we use the intervening inertial motion to estimate the position at the latest sensor time with unknown heading and no vehicle thrust model. A squared-range relaxation supplies one heading candidate. A robust fit to the original distances then corrects the integrated trajectory for a body-fixed accelerometer offset. We derive when a free quadratic can represent the same correction and which part of the motion carries heading information. With the same candidate policy, motion correction and a trained radio mean reduce horizontal endpoint root-mean-square error from 1.625 to 0.159 m on six indoor flights. Outdoors, motion correction alone reduces error from 2.930 to 1.631 m against a prism reference on ten recordings. With common constant calibration, eight-anchor horizontal error is 0.133 m, compared with 0.356 m for a range-only quadratic reconstruction. The single heading candidate matches the objective reached with four uniform starts on all 528 indoor configurations within the prescribed tolerance, using 62.1 % fewer function evaluations. As a navigation application, conditional handoff to an invariant filter restores late horizontal tracking on all six indoor flights after a 20 s outage.**

Index Terms— **Inertial navigation, invariant extended Kalman filter, observability, position acquisition, ultra-wideband ranging.**

I. Introduction

A quadrotor emerging from radio shadow receives one range at a time. Each packet constrains the antenna to a sphere at the instant of the exchange. By the time another anchor replies, the aircraft has moved. Treating these ranges as a set of simultaneous spheres introduces a motion error. Integrating the inertial measurement unit (IMU) between exchanges links the antenna positions, but an error in tilt or accelerometer bias bends the integrated path. The bend grows with the age of the oldest packet. A longer window can therefore improve the geometry of multilateration while making the assumed inertial trajectory less accurate.

Published range-only methods recover a trajectory by imposing a basis on its positions [1], while a model of quadrotor thrust and drag supplies motion information to a moving-horizon tracker [2]. We consider an alternative for an aircraft carrying one ultra-wideband (UWB) tag in a surveyed anchor field. Gravity alignment is available from an earlier IMU prefix. Absolute yaw and endpoint velocity are unknown. The desired output is the position at the most recent sensor time. Writing that fix into a running navigation filter poses a further question because velocity and attitude also determine the next predicted position.

Our approach first eliminates a polynomial translation from squared ranges and solves a two-dimensional unit-circle problem for yaw. The seed is inexpensive enough to test against several nonlinear starts. We then fit the unsquared ranges, permitting a body-fixed acceleration offset over the observed interval. A free world-frame quadratic provides a competing explanation for the same endpoint improvement. We derive the leading term shared by the two models and identify a motion for which ranges can recover the trajectory while leaving heading indistinguishable from bias.

We make three contributions. First, the endpoint formulation turns heading search into a circle problem while retaining the original metric ranges for the final estimate. Second, an explicit bias sensitivity and a conditional heading calculation explain why a longer observation interval can improve position even when yaw remains weak. Third, we quantify the value of these choices from two flight campaigns using matched range-only and vehicle-model estimators. A conditional handoff to a running invariant extended Kalman filter (InEKF) demonstrates one navigation use of the acquired position and velocity. We compare the handoff against continued filtering from the same pre-outage state.

II. Related work

A. Trajectory reconstruction and observability

Batista *et al.* obtained observability conditions and a filter for single-range navigation with measured relative velocity [3]. Theodosis *et al.* showed how quadratic outputs of certain linear systems admit a finite-dimensional linear immersion [4]. Their single-anchor navigation example takes world acceleration as an input. Both results explain how motion supplies information missing from an instantaneous range. On a quadrotor, the accelerometer reports body-frame specific force. Converting it to world acceleration already requires the heading and bias that the window is asked to estimate.

Pacholska *et al.* used an assumed trajectory basis to recover continuous position from distances alone [1]. Lifting products of basis coefficients makes the algebra linear and leads to explicit recovery conditions. Goudar *et al.* developed range-only trajectory initializers with guarantees under specified configurations of tags, beacons and motion [5]. A known baseline between body-mounted radios can supply attitude information that a single colocated tag lacks. Here, the known component is the trajectory shape obtained by integrating an IMU in a gravity-aligned frame. Separating the coefficients that enter linearly follows the variable-projection principle of Golub and Pereyra [6]. Our use of that principle leaves two coupled coefficients on a unit circle. The physically

admissible raw-range trajectories form a smaller set than those admitted by the squared-range relaxation.

B. Asynchronous fusion and inertial bias

Tightly coupled UWB/IMU estimators directly relate the radio measurement to a moving pose. Hol *et al.* connected heading estimation to acceleration excitation [7]. Kok *et al.* developed an optimization-based estimator with an asymmetric time-of-arrival error model [8]. Li *et al.* fitted continuous-time splines to asynchronous UWB and IMU samples [9]. Jiang *et al.* initialized a joint range–inertial graph and incrementally optimized it on recorded single-sided two-way ranging (TWR) data [10]. Incremental smoothing can efficiently relinearize a trajectory as new factors arrive [11]. These methods represent motion throughout a trajectory. Our window compresses the translational uncertainty into endpoint position, endpoint velocity and three acceleration-correction coefficients. This makes it possible to compare two physical explanations for the same range history at equal model dimension.

IMU preintegration incorporates the sensitivity of integrated increments to sensor biases [12]. We use that sensitivity in an endpoint-centered form. It has two consequences for acquisition. The leading position perturbation can agree with a constant world acceleration even when the true offset is body-fixed. With strong rotation, an unrestricted world acceleration and a constant body bias produce different families of positions. The heading question is consequently about which component of a change in yaw survives after fitting all the motion coefficients.

C. Radio calibration and navigation benchmarks

Radio error also varies systematically with geometry and hardware. Prorok and Martinoli learned spatial error models for indoor ranging [13]. Ledergerber and D'Andrea used maximum likelihood to calibrate systematic ranging errors [14]. Shalaby *et al.* characterized antenna delays and power-dependent bias and variance in TWR [15]. Zhao *et al.* combined learned bias correction with robust filtering for time differences of arrival (TDoA) [16]. Goudar and Schoellig studied excitation conditions for estimating the spatial and temporal offsets between a radio and an IMU [17]. Ferrero-Guillén *et al.* analyzed a further source of error arising from motion during the messages of an asynchronous TWR exchange [18]. Evaluating a finished range at its transaction time addresses motion *between* packets. It leaves motion within the exchange and unmeasured antenna delays in the observation error. We therefore compare motion models under a common radio calibration and examine transfer with its coefficients removed.

Pfeiffer *et al.* released a vehicle-model moving horizon estimator (MHE), a companion extended Kalman filter (EKF) and a quadrotor dataset [2], [19]. Their later MHE uses adaptive step sizes and switching variables for outlier handling [20]. We test the released 2021 classes at matched endpoint times and retain their dynamics and attitude handling. Oursland and Mehrabian [21] used an InEKF for a tag with a known lever arm and signed TDoA. Invariant error coordinates can improve the linearization of navigation dynamics under the hypotheses studied by Barrau and Bonnabel [22]. They cannot supply an independent heading measurement. We use the native TDoA filter on recordings with measured attitude, and assess the zero-lever-arm TWR acquisition model on separate flights.

Table I relates each estimator's output to the source of its motion information. For acquisition, the common point of comparison is the position available at a specified terminal time.

III. A yaw candidate from asynchronous ranges

A. Endpoint coordinates

Let R map body vectors to a navigation frame with vertical e_3, and write v, p for the velocity and position of the IMU. The accelerometer and gyro report f_m, ω_m with biases b_a, b_g. For nominal propagation,

$$\dot{R} = R[\omega_m - b_g]_\times, \quad \dot{v} = R(f_m - b_a) + g, \quad \dot{p} = v, \tag{1}$$

where $g = -9.81 e_3$ in m/s^2 and $[x]_\times y = x \times y$. A TWR transaction i supplies range z_i to surveyed anchor a_i at time t_i. The window takes the antenna and IMU to be colocated. With $D_i = \|p_i - a_i\|$ and $n_i = (p_i - a_i)/D_i$, its observation model is

$$z_i = D_i + \mu_{a_i}(D_i, n_i) + e_i. \tag{2}$$

Here the anchor subscript on μ identifies a radio link. The scale σ_i describes the inlier range error. The fit also accommodates occasional large residuals.

Write $R(t) = R_z(\psi)Q(t)$, where Q follows nominal gravity alignment and gyro integration in an arbitrary initial yaw frame. Let $s(t)$ be a nominal double integral of acceleration in that frame and set T to the last available sample. Defining $\tau_i = t_i - T$ gives

$$\begin{aligned} d_i &= s(t_i) - s(T) - \tau_i \dot{s}(T), \\ p_i &= p_T + \tau_i v_T + R_z(\psi) d_i. \end{aligned} \tag{3}$$

Reanchoring removes the integration constants even when the starting velocity is unknown. Both $d(T)$ and $\dot{d}(T)$ vanish. Gravity commutes with the yaw rotation, so d_i includes its displacement. The design evaluates the held IMU model at each transaction time. No simultaneous group of anchor readings is required. We call (3) the *affine* model because its free translation is affine in time. Its nominal inertial path can be highly curved.

B. Lifting and elimination of translation

For the seed, write $p_i = c(\tau_i) + R_z(\psi)d_i$, where $c(\tau) = \sum_{j=0}^{K} c_j \tau^j$. Acquisition uses $K = 1$. The general degree clarifies what information a more flexible translation removes. Subtract a constant radio offset to obtain

TABLE I

Motion assumptions and estimation targets in related range-based navigation methods.

Approach	Source of motion information	Estimated quantity and relation to acquisition
Range-only lifting [1]	A finite trajectory basis and surveyed anchors	Continuous tag position from distances. Its degree-one and degree-two forms isolate the value of IMU shape.
Range-only pose initialization [5]	Known rigid tag geometry and a motion model	Pose or trajectory under specified tag and beacon conditions. A rigid tag baseline can encode orientation.
UWB–inertial smoothing [9], [10]	IMU samples and a time-indexed spline or factor graph	Pose and inertial states over a trajectory. Joint initialization and continued smoothing share the range history.
Quadrotor MHE and EKF [2], [20]	Vehicle dynamics, including thrust and drag	Position and velocity on a rolling horizon or in a recursion. The released 2021 versions provide matched-window controls.
TDoA InEKF [21]	IMU propagation and an offset antenna	Recursive attitude, velocity and position. The lever arm couples orientation to an instantaneous range difference.
Profiled single-tag window (this work)	Gravity-aligned IMU displacement, a rotating offset and one yaw candidate	Endpoint position and velocity with free yaw. The correction requires neither a tag array nor vehicle-specific thrust parameters.

$r_i = z_i - \beta_{a_i}$ (zero offset when uncalibrated), and form

$$y_i = \tfrac{1}{2}(r_i^2 - \sigma_i^2 - \|a_i\|^2 - \|d_i\|^2) + a_{iz}d_{iz}. \tag{4}$$

In the noiseless identity take $\sigma_i = 0$. For $u = (\cos\psi, \sin\psi)^\mathsf{T}$, expansion then yields

$$y = B\nu + Dc + Cu. \tag{5}$$

The vector c stacks the c_j. The row of B contains τ_i^ℓ for $0 \le \ell \le 2K$, followed by the pairs $(\tau_i^j d_{ix}, \tau_i^j d_{iy})$ for $0 \le j \le K$. The remaining rows are

$$\begin{aligned} D_{ij} &= \tau_i^j[-a_{ix}, -a_{iy}, d_{iz} - a_{iz}], \\ C_i &= [-a_{ix}d_{ix} - a_{iy}d_{iy},\ a_{ix}d_{iy} - a_{iy}d_{ix}]. \end{aligned} \tag{6}$$

For a physical path, the coefficients in ν are constrained. The coefficient of τ^ℓ in $\|c(\tau)\|^2/2$ is $\frac{1}{2}\sum_{j+k=\ell} c_j^\mathsf{T} c_k$. The two horizontal cross-term coefficients for each j are the components of $(R_z(\psi)^\mathsf{T} c_j)_{xy}$. Releasing these identities makes ν independent of c and u. This is the only relaxation in (5).

For an ideal zero-mean Gaussian inlier, the centered half-squared range has variance $D_i^2\sigma_i^2 + \sigma_i^4/2$. We substitute $\max(r_i, \sigma_i)$ for the unknown distance in this expression and place the inverse standard deviations on the diagonal of W. With $L = W[B\ D]$, define $P = I - LL^\dagger$, $\bar{y} = PWy$ and $\bar{C} = PWC$. Minimizing over all the linear coefficients gives

$$F(\psi) = \left\|\bar{y} - \bar{C}u(\psi)\right\|^2. \tag{7}$$

This projection is exact within the relaxed squared-range model at fixed weights. The implementation removes the left singular vectors whose singular values exceed 10^{-10} times the largest value. It never needs to materialize the dense projector.

Set $H = \bar{C}^\mathsf{T}\bar{C}$, $h = \bar{C}^\mathsf{T}\bar{y}$ and $\Delta = H_{22} - H_{11}$. The terms varying with yaw are

$$\begin{aligned} F(\psi) = &-\tfrac{\Delta}{2}\cos 2\psi + H_{12}\sin 2\psi \\ &- 2h_1\cos\psi - 2h_2\sin\psi + \text{const}. \end{aligned} \tag{8}$$

Only the first two angular harmonics remain. With $q = \tan(\psi/2)$, setting the derivative to zero gives

$$\begin{aligned} 0 = &(H_{12} + h_2)q^4 + 2(h_1 - \Delta)q^3 - 6H_{12}q^2 \\ &+ 2(h_1 + \Delta)q + (H_{12} - h_2). \end{aligned} \tag{9}$$

Checking the real roots and $\psi = \pi$ covers the circle. The implementation also checks zero for a degenerate polynomial. We call the policy that retains one minimum of (7) *Profile-1*. Its global solution concerns the seed objective. The subsequent robust distance fit can have distinct local minima.

C. Identifiability and sensitivity of the heading seed

For an exact noiseless model the physical $u_\star$ makes (7) zero. If $\bar{C}$ has column rank two, every minimizer obeys $\bar{C}(u - u_\star) = 0$ and hence equals $u_\star$. Once yaw is known, translation in the lifted equations is unique if $(I - WB(WB)^\dagger)WD$ has rank $3(K+1)$. These are sufficient conditions. A rank-one projected system may intersect the circle twice, or tangentially at one point.

The row count gives a practical interpretation. For $K = 1$ the lift has seven auxiliary and six translation columns. If these 13 columns are independent, at least 15 transactions are needed to leave two projected directions. Adding a quadratic raises these counts to 11 and nine, hence 22 transactions in general position. Collecting that many packets does not guarantee useful geometry. Poor anchor height diversity or a nearly polynomial inertial shape can leave the smallest singular value of $\bar{C}$ very small. Even perfect arithmetic on (9) then gives little protection against an error in d_i.

There is also a finite perturbation bound. At fixed design let $\bar{y} = \bar{C}u_\star + \epsilon$, and let $\hat{u}$ minimize (7). Feasibility of $u_\star$ gives $\|\bar{y} - \bar{C}\hat{u}\| \le \|\epsilon\|$. The triangle inequality and the chord length on the unit circle then yield

$$\left|\sin\frac{\hat{\psi} - \psi_\star}{2}\right| \le \frac{\|\epsilon\|}{\sigma_{\min}(\bar{C})} \quad \text{if } \operatorname{rank}(\bar{C}) = 2. \tag{10}$$

The denominator can be computed before any nonlinear fit. The numerator is the departure from the physical heading in the projected equations, which can include inertial mismatch. This relation explains why concentrating the anchors in a shallow plane and adding more starts address different sources of error.

IV. Fitting the physical trajectory

A. A rotating accelerometer offset

Suppose the true accelerometer bias differs from the nominal value by a constant δb over the window. At fixed nominal rotation its contribution to world acceleration is $-R_z(\psi)Q(t)\delta b$. Integrating twice and subtracting the endpoint position and velocity contributions gives

$$B_i^a = \int_{t_i}^{T} (s - t_i)Q(s)\, ds, \tag{11}$$

$$p_i = p_T + \tau_i v_T + R_z(\psi)(d_i - B_i^a \delta b). \tag{12}$$

The sign of the kernel follows from integrating backward from T with zero terminal displacement and velocity. The integration in (11) is exact for the held rotation samples in the estimator. A changing gyro bias or an incorrect tilt changes the nominal rotation itself and can produce errors outside this three-parameter family. We interpret the fitted coefficient as a short-window motion correction.

On flights with independent radio calibration data we use the mean

$$\mu_a(D, n) = \beta_a^g + s_a(D - \bar{D}_a) + \kappa_a^{\mathsf{T}} n. \tag{13}$$

The link intercept, slope and direction coefficients are robust ridge estimates from training flights. The simpler constant-offset model uses a training median β_a. Both use the same training median-absolute-deviation range scales. A line-of-sight coefficient can absorb an apparent anchor displacement, since a small displacement δa changes distance by $-n^{\mathsf{T}}\delta a$ to first order. Its presence in (13) therefore does not identify a physical survey error. We freeze these coefficients before scoring and remove the whole mean at a site without calibration.

Define $r_i^p = (D_i + \mu_{a_i} - z_i)/\sigma_i$, and let $\rho_2(r)$ equal $r^2/2$ for $|r| \le 2$ and $2|r| - 2$ otherwise. The endpoint fit minimizes

$$\min_{p_T, v_T, \psi, \delta b} \sum_i \rho_2(r_i^p) + \frac{\|\delta b\|^2}{2\sigma_b^2}, \qquad \sigma_b = 0.10 \text{ m/s}^2. \tag{14}$$

The prior on the three bias components stays quadratic. We first fit a constant-velocity translation to the ranges from two starts, a squared-range linear solution and a stationary anchor-centroid start. Only the better translation enters (14). Profile-1 supplies its yaw and δb starts at zero. Thus the principal acquisition method performs two common translation fits and one final ten-variable distance fit. The raw fit has a cap of 120 function evaluations per start. The affine control follows the same seed policy with $\delta b = 0$.

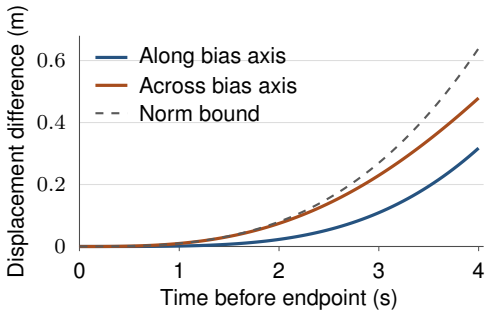


Fig. 1. Difference between a rotating bias correction and its leading world-frame quadratic in an analytic planar turn. The bias magnitude is 0.10 m/s^2, the yaw rate is 0.60 rad/s and $Q(T) = I$. Magnitudes along and across the endpoint bias axis are shown. The dashed line bounds the norm in (16) after multiplication by $\|\delta b\|$.

Differentiating the radio mean matters when the aircraft crosses a learned direction gradient. Write $\ell_i = R_z(\psi)(d_i - B_i^a \delta b)$. The two factors of the residual Jacobian are

$$\begin{aligned} g_i &= (1 + s_a)n_i + (I - n_i n_i^{\mathsf{T}})\kappa_a / D_i, \\ G_i &= [\, I,\ \tau_i I,\ e_3 \times \ell_i,\ -R_z(\psi)B_i^a \,], \\ J_i &= \sigma_i^{-1} g_i^{\mathsf{T}} G_i. \end{aligned} \tag{15}$$

For the constant radio mean $g_i = n_i$. The second term of g_i acts perpendicular to the line of sight and accounts for the change in predicted direction when the fitted position moves.

B. Local equivalence to a quadratic correction

Let $h_i = T - t_i$. A freely fitted world acceleration c_W adds $\frac{1}{2}h_i^2 c_W$ to (3). It costs three parameters, the same number as δb, and has no prior. For a fixed attitude the two motion models span exactly the same corrections through $c_W = -R_z(\psi)Q(T)\delta b$. When $\|\dot{Q}\|_2 \le \Omega$, the difference between their leading terms obeys

$$\left\| B_i^a - \tfrac{1}{2}h_i^2 Q(T) \right\|_2 \le \int_0^{h_i} (h_i - u)\Omega u\, du = \tfrac{1}{6}\Omega h_i^3. \tag{16}$$

If Q is twice continuously differentiable, expanding the integrand in (11) gives $B_i^a = \frac{1}{2}h_i^2 Q(T) - \frac{1}{6}h_i^3 \dot{Q}(T) + O(h_i^4)$. The cubic term is the first shape difference caused by rotation. A short or weakly turning flight may therefore reveal the need to correct acceleration long before it distinguishes a body-fixed offset from a world-frame one. Fig. 1 gives an example with prescribed rotation so that the two components of this difference can be read in metres.

C. Heading sensitivity after fitting nuisance states

For local analysis, multiply each radio Jacobian row in (15) by $\sqrt{\min(1, 2/|r_i^p|)}$, taking weight one at zero. Partition the result into the yaw column $\tilde{j}_\psi$ and the position, velocity and correction columns $\tilde{J}_n$. To first order, the part of a yaw change that these nuisance

variables cannot absorb has squared norm

$$\mathcal{I}_{\psi|n} = \min_a \left\|\tilde{j}_\psi - \tilde{J}_n a\right\|^2 = \left\|(I - \tilde{J}_n \tilde{J}_n^\dagger)\tilde{j}_\psi\right\|^2. \quad (17)$$

This is a local Gauss–Newton curvature with fixed robust weights. Adding the bias prior penalizes the minimizing nuisance motion and can increase this number without adding information from the ranges. A locally sharp minimum can also coexist with a distant yaw mode.

A limiting case exhibits exact ambiguity. Assume constant Q, constant measured specific force f_m, zero lever arm and a freely fitted constant bias. For any second yaw ψ', choose

$$b'_a = f_m - Q^\mathsf{T} R_z(\psi')^\mathsf{T} R_z(\psi) Q(f_m - b_a). \quad (18)$$

Both pairs (ψ, b_a) and (ψ', b'_a) generate the same world acceleration. With the same starting position and velocity, their positions, radio directions and all ranges coincide even for ideal anchors. A bias prior can select between them. The radio data cannot. A time-varying f_m generally makes the right side of (18) time dependent, so constant attitude alone is insufficient for this ambiguity.

A related loss occurs with free polynomial translation. Adding a vector polynomial $b(\tau)$ of degree at most K to d and replacing $c(\tau)$ by $c(\tau) - R_z(\psi)b(\tau)$ preserves every position for each fixed yaw. If d lies entirely in this polynomial space, varying yaw cannot change the best range fit. The bias model in (14) restricts its allowed correction through B_i^a and a prior.

V. Flight data and comparison protocol

A. Acquisition windows and reference positions

The Delft dataset [19] contains two TWR recordings for each of six motions. We use the earlier recording of each motion to fit radio calibration and the later one to score acquisition. All six evaluation flights share the room and campaign date and were available during method development. A held-motion calibration control excludes the earlier recording of the target motion as well. It tests whether a directional calibration generalizes to a different motion in the same anchor installation.

The retained clock starts after approximately 1.5 s of initial IMU data used for tilt and nominal bias. A common offline 0.20 s offset aligns UWB and IMU time with OptiTrack scoring time. The valid interval is selected with reference support, and the reference is interpolated at the actual last available sample. The window never receives a test position. At 5 s we evaluate 2 and 4 s windows. At 10, 20 and 30 s we also evaluate an 8 s window. This gives 11 segments per flight and four distinct endpoint times. All eight anchors and the four lowest anchors form two nested layouts. Thus there are 66 physical windows, 132 layout-specific windows and only 24 distinct flight/endpoint times.

For the seed comparison we also solve in four equivalent inertial yaw coordinate frames. The geometry and data are unchanged, although a finite-budget optimizer can reach different minima. The 528 fits characterize solver behavior across coordinates. We additionally fix one coordinate frame when comparing flights so that a flight with more yaw parameterizations receives no extra evidential weight.

The outdoor QDrone dataset [23] contributes ten recordings from a field, a building, a bridge and a tunnel. Nested 2, 4 and 8 s windows end at the midpoint of sensor coverage in each recording, giving 30 segments at ten endpoint instants. The published total-station prism position serves as a proxy because its exact transform to the antenna is unavailable. Both fits use the same IMU prefix, zero radio mean and a 0.30 m radio scale. We use the registered coordinates directly, without site-specific radio fitting or truth-based realignment.

B. Metrics and matched inputs

For window i, let $e_{i,xy} = \left\|\hat{p}_{i,xy} - p^*_{i,xy}\right\|$. For an explicitly specified set S we report

$$E_{xy}(S) = \sqrt{|S|^{-1} \sum_{i \in S} e_{i,xy}^2}. \quad (19)$$

Vertical (Z) and three-dimensional (3D) errors use the corresponding components of the position residual. Within the same set $E_{3\mathrm{D}}^2 = E_{xy}^2 + E_z^2$, which allows a vertical failure to be distinguished from a horizontal one. We use flight-level pairs to show heterogeneity and retain the larger set of fits where measuring optimization reliability is the purpose. The reported summaries are descriptive. Overlapping windows are correlated, as are alternative layouts of the same anchor field.

For comparisons with released Delft estimators, all methods receive the same changed positive range packets, surveyed anchors, window boundaries and reference endpoint. The two shared conditions are raw ranges with a fixed 0.25 m scale, and ranges corrected by training-only constant offsets with training scales. The distance-dependent mean is reserved for the model ablation. Each method restarts for each external 2, 4 or 8 s window. The MHE retains its internal 20-node horizon at 50 Hz, one gradient step and its sample-consensus rejection. The EKF runs at 100 Hz. Both retain their quadrotor thrust/drag model, using a thrust proxy from measured acceleration, and start from the authors' zero translation and level zero-yaw attitude. We disable the auxiliary height input and allow any final pending packets to be consumed. The MHE gradient does not use the supplied radio variances. Our fit inherits tilt and gyro history from the prefix and searches yaw. The comparison matches radio information and endpoint times while keeping the estimators' distinct attitude and vehicle assumptions.

For continuous tracking we compute a time-domain XY RMSE from the most recently emitted state on a 20 Hz grid within each flight and interval. Section VII reports medians over reference-supported flights and paired differences between policies.

TABLE II

Endpoint RMSE in metres with one profiled yaw candidate per fit. Delft pools all four yaw coordinates. QDrone uses the prism proxy.

	Affine			Body bias		
Population	XY	Z	3D	XY	Z	3D
Delft, eight	1.585	.758	1.757	.116	.209	.239
Delft, low four	1.664	1.163	2.030	.192	.526	.560
Delft, pooled	1.625	.981	1.898	.159	.400	.431
QDrone	2.930	1.655	3.365	1.631	1.779	2.414

TABLE III

Delft control for motion and radio calibration. Four uniform yaw starts, reference coordinate and 66 physical windows per layout. Entries are RMSE in metres.

		Eight anchors		Low four	
Motion	Radio	XY	3D	XY	3D
Affine	Constant	1.631	1.799	1.665	2.029
Quadratic	Constant	.127	.320	.201	.924
Body bias	Constant	.133	.279	.203	.874
Affine	Directional	1.654	1.756	1.653	1.924
Quadratic	Directional	.111	.273	.188	.591
Body bias	Directional	.116	.239	.192	.560

VI. Endpoint accuracy and competing estimators

A. Effects of trajectory correction and radio calibration

Table II compares the affine and bias-corrected fits under the same single-candidate policy. In Delft the headline comparison changes both the motion correction and the radio mean. In QDrone the mean is zero for both fits. Separating the two changes is essential, since both can reduce a systematic range residual.

Across Delft configurations, horizontal RMSE falls by 90.2%. The bias-corrected fit improves 440 of the 528 paired horizontal errors. All 144 configurations with 8 s of data improve, and the largest change is at that duration. Pooled XY for the 2, 4 and 8 s windows changes from 0.222, 0.709, 2.991 m to 0.164, 0.175, 0.126 m, respectively. Keeping only the reference coordinate gives 1.645 and 0.159 m over 132 layout-specific windows. All six flight-level horizontal RMSEs decrease in that frame (Fig. 2). The difference between 1.645 and 1.625 m is a finite-optimization effect in the affine fit when equivalent yaw coordinates are used.

On QDrone, the motion correction reduces pooled horizontal error by 44.3% with no site calibration. At 2, 4 and 8 s the paired values are 1.634 to 1.659 m, 2.120 to 1.629 m and 4.312 to 1.605 m. Pooling the three lengths gives a horizontal improvement in every recording. Unlike Delft, which uses motion capture, this result measures distance to a registered prism. The horizontal and vertical columns in Table II preserve that distinction in the interpretation of 3D accuracy.

Table III holds the raw model's four-start policy constant across motion families. A directional radio mean alone leaves the affine fit above 1.6 m in XY for both layouts. Under constant calibration, either three-parameter acceleration correction removes most of that error. With directional calibration the free quadratic and body bias give pooled XY of 0.155 and 0.159 m, respectively. Each wins on three flights. The corresponding 3D values are 0.460 and 0.431 m. These outcomes agree with the leading-order equivalence in (16). They give little basis for interpreting the fitted coefficient as a measured body bias. On QDrone the four-start quadratic and body models give 1.677 and 1.631 m in XY, and 2.352 and 2.414 m in 3D. Even the ordering depends on which position components matter to the application.

TABLE IV

Delft seed policies on the same 528 configurations and body-bias objective. Evaluations include both common translation fits. Capped final solutions are retained.

Yaw policy	Misses	Evaluations	XY (m)	3D (m)
One local yaw	7	36,313	.154	.429
Profile-1	0	32,789	.159	.431
Four uniform yaws	0	86,508	.159	.431

Calibration adds a benefit that survives holding out the target motion from training. Changing the body-model mean from constant to directional changes XY from 0.135 to 0.119 m with eight anchors and from 0.204 to 0.199 m with the four lowest. The corresponding 3D changes are 0.289 to 0.250 m and 0.864 to 0.616 m. The larger 3D gain with low anchors reflects the sensitivity of height to the range mean.

B. Cost and reliability of the yaw candidate

We next fix the body model and compare its observed objective for a local yaw, Profile-1 and four uniform yaw starts. Let J denote the stored objective in twice the normalization of (14). A miss on a real window means $J - J_{\min} > 10^{-4}(1 + J_{\min})$, where $J_{\min}$ is the best among these three policies for that same window. Raw objectives across different motion families have different meanings and are never used for this count.

Profile-1 reaches the best observed objective within tolerance everywhere in Table IV, using 62.1% fewer function evaluations than four starts. A single local yaw misses seven times despite giving similar physical endpoint errors. On the outdoor recordings the miss counts for local, profiled and four-start fits are 1, 0 and 0. Switching the Delft Profile-1 *motion model* from affine to body bias uses 32,632 versus 32,789 evaluations, an increase of 0.48%. The evaluation count includes both translation fits. Per-evaluation work varies with the number of fitted variables.

A controlled simulation isolates the numerical seed from model mismatch. Eight anchors occupy $(\pm 4, \pm 4, z)$ m with $z \in \{0, h\}$. The endpoint is 30 m from their centroid horizontally. Each of 200 windows per height span contains an excited four-second inertial shape, 240

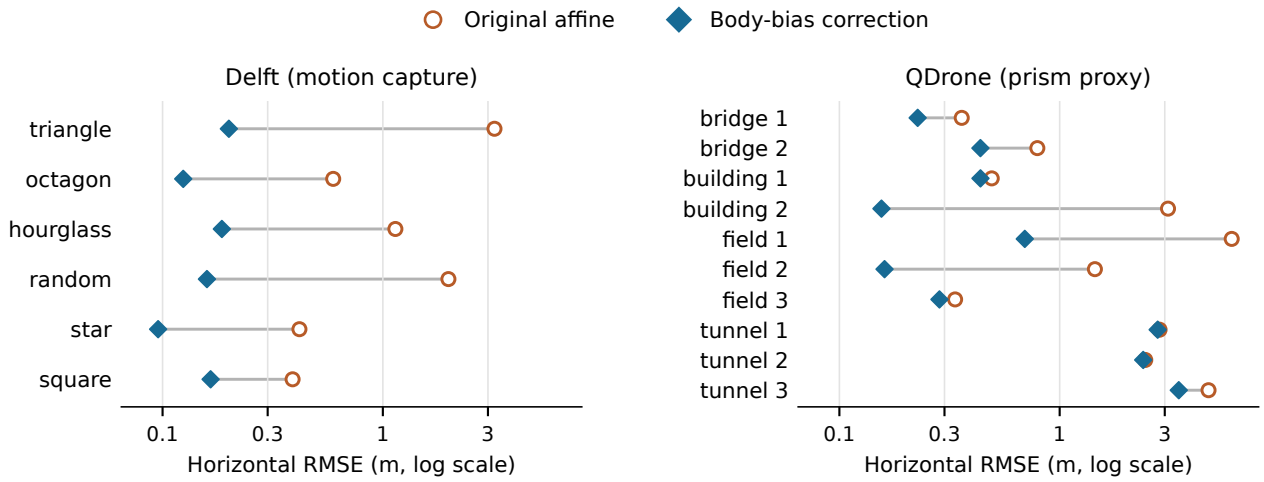


Fig. 2. Horizontal endpoint RMSE within each recording. Each pair uses the same single yaw candidate policy. Delft pools 11 windows under two layouts at one fixed yaw coordinate, giving 22 errors per flight. QDrone pools three window lengths at one midpoint per recording. The horizontal axis is logarithmic. Each line represents one recording.

asynchronous transactions and 0.10 m Gaussian range noise. This experiment refines the *affine* model. Its reference is the best of five policies, local, Profile-1 and 2, 4 or 12 uniform yaws, with objective tolerance $\max(10^{-3}, 10^{-5}|J_{\min}|)$. At $h = 3$ m, Profile-1 and two uniform starts each have zero misses and 0.509 m 95th-percentile 3D error. A local start misses 41 times. In 40 serial timing measurements per policy and geometry on an Intel Core i5-12600KF with one BLAS thread, median times at this height are 47.1 ms for Profile-1 and 66.6 ms for two starts. At $h = 0.1$ m, the corresponding two policies miss 12 and 10 times. With four and twelve starts that count falls to four and one. The 95th-percentile 3D error for Profile-1 reaches 3.946 m, dominated by height. The height span governs the physical precision attainable at the best observed fit.

C. Range-only and vehicle-model estimators

The relaxation of Pacholska *et al.* [1] reconstructs motion using only the ranges. We evaluate the published line basis $(1, t)$ and a prespecified quadratic $(1, t, t^2)$ in the same solver. The latter represents the entire path, whereas the quadratic in Table III corrects a measured inertial path. Under common constant calibration, incorporating the IMU and fitting the body offset reduces horizontal RMSE from 0.356 to 0.133 m with eight anchors and from 0.474 to 0.204 m with the four lowest (Table V). On the low layout the corresponding vertical errors are 7.982 and 0.854 m. The range-only linear system has full numerical rank in every window. The inertial shape supplies additional structure when shallow sight lines carry little independent height information.

Fig. 3 shows where the improvement occurs. On the eight-anchor layout, the horizontal error of the range-only quadratic rises from 0.142 m at 2 s to 0.591 m at 8 s. The inertial fit gives 0.121 and 0.113 m at those durations. With the four lowest anchors the corresponding 8 s errors are 0.685 and 0.179 m. The additional IMU input allows longer segments to retain measured turns, while a range-only polynomial must explain the entire maneuver using the same few coefficients.

The Pfeiffer trackers [2] supply a complementary point of comparison. With eight anchors and constant correction, the single-candidate method reaches 0.279 m in 3D, between the author's EKF (0.254 m) and MHE (0.343 m). Of 66 endpoints, 63 are within 0.5 m for our fit, compared with 65 for the EKF and 56 for the MHE. Both vehicle estimators achieve lower pooled XY in the shared radio conditions. The window method reaches this range of accuracy using freely estimated yaw, with no thrust, mass or drag parameters. All finite outputs, including evaluation-capped fits, enter the comparisons.

VII. From a position fix to filter recovery

A. One checkpoint and one use of each range

The live TWR filter propagates $x = (R, v, p, b_g, b_a)$ with a full 15-dimensional right-invariant-error covariance. The window and the filter may have used the same ranges. We resolve this overlap by recomputing one conditional interval posterior from the pre-outage checkpoint, using the acquired motion as a linearization trajectory.

At startup or when the first range arrives after a gap longer than 1 s, the live filter continues emitting its usual predictions and corrections. After 2 s of new sensor time we attempt recovery once if there are at least 16 positive packets and four distinct anchors. The checkpoint c is the last complete posterior before the outage. The interval includes the IMU throughout the gap and the unique range packets after the checkpoint up to time k. At startup the initial prior plays the same role. The seed uses only post-

TABLE V

Published-method endpoint comparison on Delft. Each entry is an RMSE in metres over 66 windows per layout. Constant and raw rows are separate shared-input comparisons. The single Profile-1 row uses the body model.

		Eight anchors			Four lowest anchors		
Estimator and configuration	Radio	XY	Z	3D	XY	Z	3D
Pacholska *et al.*, degree one, unweighted	Constant	.669	.425	.793	.689	5.803	5.844
Pacholska *et al.*, degree one, author weights	Constant	.672	.411	.788	.676	6.268	6.305
Range-only degree two, author weights	Constant	.356	.491	.606	.474	7.982	7.996
Pfeiffer *et al.*, author MHE	Constant	.121	.321	.343	.187	.876	.895
Pfeiffer *et al.*, author EKF	Constant	.105	.231	.254	.188	.869	.889
Single Profile-1 with body bias	Constant	.133	.245	.279	.204	.854	.878
Pfeiffer *et al.*, author MHE	Raw	.150	.549	.569	.219	.235	.321
Pfeiffer *et al.*, author EKF	Raw	.172	.506	.534	.229	.243	.334
Single Profile-1 with body bias	Raw	.205	.523	.562	.397	.241	.465

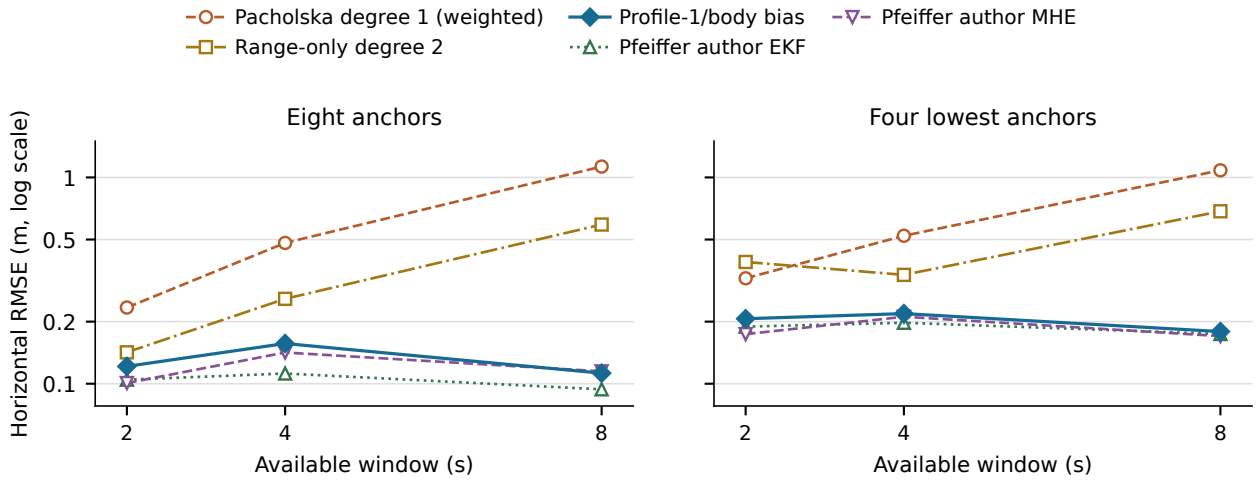


Fig. 3. Horizontal endpoint RMSE versus window length under identical constant range calibration. Each layout contributes 24 two-second, 24 four-second and 18 eight-second windows. The range-only curves use the published weighting. The two quadrotor trackers use the authors' classes. Duration is the available data interval for all methods.

return ranges for recovery, while the prior and the interval dynamics preserve the earlier information.

Let P_c denote the checkpoint covariance in the appropriate local chart, u_j the IMU sample and S_w the covariance of the native discrete process noise. In a local state difference $\boxminus$, the interval problem has the form

$$\begin{aligned}\min_{x_{c:k},w_{c:k-1}} \quad & \tfrac{1}{2}\,\|x_c \boxminus \hat{x}_c\|^2_{P_c^{-1}} + \tfrac{1}{2}\sum_{j=c}^{k-1}\|w_j\|^2_{S_w^{-1}} \\ & + \sum_{i\in\mathcal{E}_{c:k}} \rho_2\!\left(\frac{\|p_i - a_i\| + \beta_{a_i} - z_i}{\sigma_i}\right), \qquad (20)\\ & x_{j+1} = f_j(x_j, u_j, w_j).\end{aligned}$$

Here $\|z\|^2_M = z^{\mathsf{T}} M z$ and $\mathcal{E}_{c:k}$ excludes packets already in the checkpoint. Each smoother pass starts from the original prior and the same unique factors. A profiled affine fit followed by the body-bias refinement initializes the reference trajectory. This use of an affine fit before refinement is specific to the smoother seed. Neither its endpoint nor the bias penalty from (14) is an additional observation in (20). The iterated extended Kalman smoother makes four passes using native discrete inertial dynamics and bias random walks. It robustly downweights ranges within the batch, whereas the continuing native filter retains its innovation gate.

At handoff, we discard the provisional terminal posterior for the interval and take the full smoothed state and covariance. Denote the smoother's local error by $\delta x_P = (\delta\theta_b, \delta v_W, \delta p_W, \delta b_g, \delta b_a)$. After the final attitude retraction its covariance is reset by $G = \mathrm{diag}(J_r(\delta\theta_b), I_{12})$ with the right Jacobian J_r of $SO(3)$. For left injection into the right-invariant filter the coordinate map at the new endpoint is

$$A(X) = \begin{bmatrix} R & 0 & 0 & 0 \\ [v]_\times R & I & 0 & 0 \\ [p]_\times R & 0 & I & 0 \\ 0 & 0 & 0 & I_6 \end{bmatrix}, \qquad P_I = A\,G P_{\mathrm{sm}} G^{\mathsf{T}} A^{\mathsf{T}}. \qquad (21)$$

This transports both navigation–bias and attitude–translation cross terms. For example, a body attitude perturbation becomes $R\delta\theta_b$ in world coordinates and contributes $[p]_\times R\delta\theta_b$ to the invariant translation error. The continuing filter therefore inherits the position–attitude coupling at the recovered state.

TABLE VI

Late horizontal tracking after range return. Entries are medians of per-flight XY RMSE (m) during $[6, 11)$ s. H continues the filter, W applies the window, and L uses lifted capture.

	Delft (6 flights)			QDrone (8 recordings)		
Gap (s)	H	W	L	H	W	L
2	.094	.093	.094	.551	.551	.551
5	.093	.093	.093	.545	.545	.545
10	.095	.092	.092	.635	.547	.544
20	39.429	.092	.092	.577	.533	.536

B. Same-history intervention on two TWR datasets

We suppress ranges for 2, 5, 10 or 20 s, ending at $t = 25$ s, keeping the IMU active. In the control H, profiled startup is followed by native recovery. In W, the identical startup is followed by conditional replacement on radio return. A third arm L uses an earlier multi-anchor lifted position/velocity capture followed by an InEKF. This project baseline supplies a practical alternative with a different history. The within-flight difference $W - H$ isolates intervention at recovery. A direct comparison between W and a filter that had already failed at startup would confound the two decisions.

The intervention produces 64 flight/gap pairs on 16 recordings. H and W agree exactly in state, biases and covariance diagonals through the two-second acquisition wait in all 64 pairs. At each gap, six indoor and eight outdoor recordings meet the 95% reference-coverage requirement for the late interval. Four of 40 outdoor recovery attempts lack the required anchor support and continue with the native filter. The analysis retains every estimate on supported reference intervals.

With a 20 s indoor outage, replacement improves all six late errors over H (Table VI). The median of the paired changes is -39.352 m. Recovery is already visible in the 2–5 s interval, where the median falls from 16.451 to 0.109 m. By 6–11 s the filter reaches the 0.092 m median of the lifted-capture benchmark. With shorter indoor gaps the native filter is already close to that accuracy, leaving little room for improvement. The useful intervention in this campaign is a restart after substantial inertial drift.

The outdoor trial tests the same intervention across sites. After a 20 s gap the two late cohort medians are 0.577 and 0.533 m, with a median within-recording change of $+0.00056$ m. At 2 s, continuing the filter gives the lower late error on six of eight supported recordings. These paired results favor using the outdoor window first as a position fix.

C. Physical heading with the original TDoA InEKF

For the original filter of Oursland and Mehrabian [21], the published antenna offset ℓ enters the ideal signed observation for anchors a, b as

$$h_{ab}(R, p) = \|p + R\ell - b\| - \|p + R\ell - a\| . \qquad (22)$$

This observation can change instantaneously with orientation through the lever arm. In the zero-lever-arm TWR model (2), heading affects distances through the trajectory history. We therefore assess physical yaw with the original InEKF class on Toronto UTIL flights with motion-capture attitude [24]. Each cohort comprises four recordings and two initial-yaw interventions (0° and 180°). The original event queue and covariance are retained. We suppress TDoA for $[10, 20)$ s and score wrapped yaw and XY on $[26, 45)$ s.

Reserving packets for an early XY update changes median horizontal/yaw RMSE from 0.112 m/5.93° to 0.087 m/6.80° in the primary cohort, and from 0.0705 m/7.22° to 0.0709 m/6.22° in the complementary cohort. The position and heading scores move independently even with genuine attitude truth. A component-wise decision at handoff is therefore consistent both with the measurement geometry in (22) and with the flight results.

VIII. Discussion and conclusion

The window recovers position by combining two complementary descriptions of motion. The IMU records turns that a low-order range-only polynomial smooths away. Three acceleration-correction coefficients let that inertial shape adapt as integration error grows. Profiling yaw makes the combination practical with one nonlinear start. Across six indoor flights and ten outdoor recordings, the gain in horizontal endpoint accuracy is largest on the longest windows. With eight indoor anchors and the same range calibration, the 8 s horizontal RMSE is 0.591 m for range-only quadratic reconstruction and 0.113 m for the proposed method. The rotating-bias derivation explains why a free quadratic also works well on short segments and how turning separates the two models.

The smaller projected singular value bounds seed sensitivity before fitting. The conditional curvature then measures the yaw information retained after refitting position, velocity and bias.

The acquisition equations require no vehicle-specific dynamics and can initialize a single InEKF. In the long indoor outage case, relinearizing one interval from its checkpoint restored the accuracy of an established capture method. The next step is to validate a component-wise handoff at a held-out site with synchronized antenna, prism and heading references. Measuring delivery latency on the flight processor and testing heading acceptance against those references would extend the present position result to a deployable recovery policy.

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